Kurtis William Ramsey

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EDUCATION

San Diego State University, College of Engineering **May 2026**
B.S. in Mechanical Engineering San Diego, CA

SKILLS

Software: MATLAB, SolidWorks, SolidCAM, Arduino IDE, Simulink, Microsoft Office, Java, TracerDAQ, Raspberrypi

Machining: HAAS CNC Mill, HAAS CNC Lathe, CNC Waterjet Cutter, MIG Welder, Flash Sintering Furnace, SHS Reactor, Bambu 3D Printers, Drill Press

Certifications: RCF Certification (CITI Programs)

EXPERIENCE

Lab Researcher **August 2025 - May 2026**
SDSU Advanced Materials Processing Lab (AMPL) San Diego, CA

- Reviewed literature on closed-cell Al foams and powder metallurgy to guide research on electrically driven spark plasma foaming of aluminum metal compacts, investigating the effect of applied current on heating rate and foam structure as an alternative to furnace-based processing
- Ran systematic trials combining TiH_2 and Al powders across 250-1000A current outputs; identified 750A as producing the fastest stable heating rate (~994°C/min), while determining reduced plateau dwell time was needed to prevent pore coalescence
- Designed a Type-C thermocouple measurement setup integrated with Omega DAQ hardware and TracerDAQ software for real-time thermal and electrical data acquisition
- Characterized the effect of green compact relative density (92-99%) and height-to-width ratio on heating behavior, identifying H/W ratio as the dominant geometric driver of heating rate
- Correlated relative density with the green compacts', proposing resistivity as a non-destructive screening method to predict foam quality before processing
- Performed Archimedes density analysis and cross-sectional porosity characterization on foamed samples to evaluate closed-cell structure quality and identify optimal processing parameters

Lead Engineer, Lunar Compatible Nomex-Nylon Recycling System **August 2025 - May 2026**
NASA LUNARECYCLE CHALLENGE San Diego, CA

- Co-designed a closed-loop, in-situ polymer recycling system converting lunar waste (Nomex, multilayer nylon pouches) into 3D-printable filament with a 5-engineer team
- Built a dual-shredder subsystem (custom grater-style + repurposed COTS shredder) reducing Nomex fabric to 14mm x 0.1mm particulate
- Developed a melt-filtration process (band-heated barrel, stainless steel mesh, compressive loading) achieving 45.4% nylon reclamation across 17 test iterations
- Designed a single-screw filament extruder with a 3-zone heater profile (120°C/170°C/200°C) to process reclaimed material at 500g/60min
- Built a networked control system (Raspberry Pi, CAN bus, I2C sensing) with remote dashboard and integrated E-stop/interlock safety features
- Machined and fabricated hardware via CNC milling, CNC lathe, CNC waterjet cutting, MIG welding, and 3D printing
- Managed requirements, a 15-item risk register, and an $8K budget; verified all system requirements via test and demonstration
- Selected as 1 of 10 finalist teams in NASA's LunaRecycle Challenge Milestone Round, earning $50,000 in funding
- Presented a Poster Presentation at the 2026 SDSU Student Symposium for research framing and technical reasoning
- Presented a poster at 2026 LSIC Spring Meeting detailing system performance and metrics

ACTIVITIES, AFFILIATIONS, AWARDS

Honors/Awards: Eagle Scout

Leadership/Projects: Eagle Scout Project - Design and Fabrication of a Collapsible Gaga Ball Pit and Transport Cart for Torrey Pines Church

Additional Experience: Server, Piazza 1909 (2021- Present) - Progressed through roles of Busser, Food Runner, and Server